Face Occlusion Estimation: A Deep Learning Approach with Classical CV Baseline

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Introduction

The Telecom Paris–Idemia Face Occlusion Challenge frames face occlusion estimation as a regression task: given a 224 × 224 pixel aligned face crop, predict the fraction of the face that is occluded, $\hat{y} \in [0, 1]$. The official evaluation metric is a **gender-balanced, occlusion-weighted mean squared error**:

$$\mathcal{L} = \frac{E_F + E_M}{2} + |E_F - E_M|$$

where for each gender group $g \in \{F, M\}$:

$$E_g = \frac{\sum_i w_i (\hat{y}_i - y_i)^2}{\sum_i w_i}, \quad w_i = \frac{1}{30} + y_i$$

The additive weight w_i makes high-occlusion samples up to 30× more influential than clear faces, explicitly penalising **false negatives** on heavily occluded faces. The imbalance term $|E_F - E_M|$ further requires that the model performs equitably across genders. A solution that excels on one group at the expense of the other is harshly penalised.

Proposed Deep Learning Solution

Architecture

Our model is a lightweight **transfer-learning regressor** built on top of a pre-trained **EfficientNet-B4** backbone [1] extracted via the `timm` library. EfficientNet-B4 was chosen for its favourable accuracy-to-parameter ratio at 224 × 224 input resolution and the depth of its compound-scaled feature hierarchy, which captures both low-level texture (essential for detecting fabric/hair occlusions) and high-level semantic structure (essential for partial face parsing).

The backbone is used as a fixed-geometry feature extractor (all layers unfrozen for fine-tuning). Its output feature vector (1792-d) is passed through a **custom regression head**:

$$\begin{gathered}\text{Linear}(1792 \to 512) \to \text{ReLU} \to \text{Dropout}(0.3)\\ \to \text{Linear}(512 \to 128) \to \text{ReLU} \to \text{Dropout}(0.3)\\ \to \text{Linear}(128 \to 1) \to \text{Sigmoid}\end{gathered}$$

The terminal Sigmoid constrains predictions to $[0, 1]$ and naturally models the bounded regression target without additional output clipping.

Data Augmentation

All training images are processed with ImageNet-standard normalisation (mean $\mu = (0.485, 0.456, 0.406)$, std $\sigma = (0.229, 0.224, 0.225)$). During training, the following stochastic augmentations are applied:

- **Random horizontal flip** (p = 0.5): face symmetry invariance.
- **Color jitter**: brightness ±0.2, contrast ±0.2, saturation ±0.1 — addresses variable lighting across identities and environments.
- **Random rotation** ±15°: robustness to slight head tilt.
- **Gaussian blur** (kernel 3 × 3, p = 0.3): simulates sensor blur and compression artefacts common in low-quality crops.

Validation inference uses the normalisation transform only (no augmentation). Test-time augmentation (TTA) is supported at inference: k passes with training augmentations are averaged to reduce prediction variance.

Training Methodology

Training uses 80 000 labelled images from the 100K crop dataset, with a stratified 80/20 train–validation split. Stratification keys are formed from the cross-product of gender and occlusion quintile to guarantee that both group distributions and target density are preserved in each split.

Optimiser: AdamW [2] with learning rate $\eta = 10^{-4}$ and weight decay $\lambda = 10^{-4}$.

Scheduler: Cosine annealing over 40 epochs with no warmup, decaying to $\eta_{\min} = 0$ to allow fine-grained convergence near the end of training.

Weighted sampler: To counteract the heavy class imbalance toward low-occlusion faces, a `WeightedRandomSampler` assigns each sample a draw probability proportional to $\frac{1}{30} + y_i$, mirroring the evaluation metric. An additional **male boost** factor of 3.0 is applied to male samples to force the model to reduce $|E_F - E_M|$. This yields a training distribution that

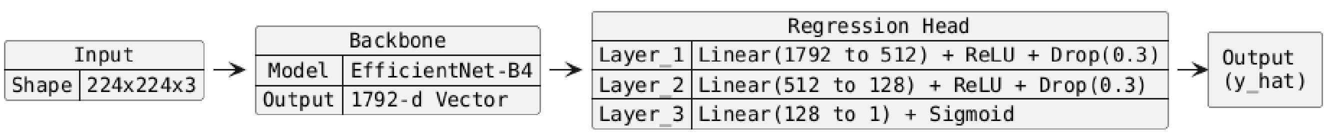

Figure 1: Condensed architectural pipeline: EfficientNet-B4 feature extraction followed by a 3-layer regression head.

is simultaneously occlusion-heavy and gender-balanced, directly targeting the structure of the loss.

Custom Loss Function

The model is trained by directly **optimising the challenge metric** rather than a surrogate. The training loss replicates $\mathcal{L}$ exactly:

$$\mathcal{L}_{\text{train}} = \frac{E_F + E_M}{2} + |E_F - E_M|$$

with per-sample weights $w_i = \frac{1}{30} + y_i$ computed inside each forward pass. In the rare case of a mono-gender mini-batch (mitigated by the weighted sampler), the loss falls back to the global weighted MSE over the batch. This end-to-end alignment between training objective and evaluation criterion removes the optimisation gap that would otherwise arise from a vanilla MSE or Huber loss.

Ensemble Inference

Final predictions are produced by averaging the outputs of **three independent EfficientNet-B4 checkpoints** (seeds 123, 42, 456) trained with identical hyperparameters. Ensemble diversity is achieved through different random initialisations and data shuffling. Each checkpoint optionally applies TTA before the predictions are averaged, further reducing variance. The ensemble prediction for sample i is:

$$\hat{y}_i = \frac{1}{K}\sum_{k=1}^{K} \mathrm{TTA}\left(\hat{y}_i^{(k)}\right)$$

Alternative Baseline Explored

Prior to adopting the deep learning approach, we developed a **fully training-free, deterministic classical CV heuristic** to establish a performance ceiling for analytical methods. The pipeline operates as follows: (1) CLAHE is applied to the Y channel of YCrCb to normalise illumination; (2) a primary skin mask is generated from fixed YCrCb chrominance ranges (Cr $\in [133, 173]$, Cb $\in [77, 127]$), which are illumination-invariant across the human skin-tone gamut; (3) an adaptive HSV secondary mask supplements the primary detector; (4) a Laplacian texture filter with anatomy-aware exclusion masks for eyes and mouth suppresses false edges from natural facial features; (5) the occluded fraction is estimated from the non-skin pixel ratio inside a central circular ROI. While this approach is **fully generalised** — requiring no training data and operating on any face crop without prior calibration — it encountered a hard performance ceiling (0.023 on the challenge metric) due to JPEG compression artefacts, low resolution, and the high semantic complexity of occlusion types (scarves, hands, hair). This validated our decision to pivot to deep learning, which can model the semantic richness of the task at the cost of data dependency.

Limitations

Data dependency: Unlike the classical baseline, our EfficientNet regressor requires tens of thousands of labelled examples. Performance degrades on out-of-distribution face types or occlusion styles not represented in the training corpus.

Computational cost: EfficientNet-B4 inference requires a GPU for practical throughput. The training pipeline (40 epochs $\times$ 80K images $\times$ 3 seeds) requires several hours on a modern NVIDIA GPU, limiting the number of hyperparameter configurations that can be evaluated.

Hyperparameter sensitivity: The gender balance and male boost parameters of the weighted sampler interact non-trivially with the loss imbalance term. The chosen value (3.0) was determined empirically and may not generalise to data with different gender ratios.

Ensemble correlation: The three checkpoints share the same backbone and architecture. Their predictions are correlated, reducing the variance-reduction benefit compared to a diverse ensemble (e.g., mixing EfficientNet-B4 with ConvNeXt or ViT architectures). We explored ConvNeXt-Base and ViT-Base variants (configurations available in `model-training/configs/`) but EfficientNet-B4 produced the best single-model scores within our compute budget.

Conclusion

We presented a deep learning solution based on fine-tuned EfficientNet-B4 that directly optimises the gender-balanced, occlusion-weighted evaluation metric. Key design decisions — weighted sampling, a custom loss identical to the metric, and multi-seed ensembling — each target a distinct aspect of the challenge's evaluation structure. A deterministic classical CV baseline was built and benchmarked first, validating the necessity of the learned approach for handling the semantic complexity of face occlusion in compressed, low-resolution crops.

Bibliography

[1] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pp. 6105–6114, 2019.

[2] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," *International Conference on Learning Representations (ICLR)*, 2019.